

ABERDARE TECHNICAL COLLEGE

MODULE 4: DATABASE DESIGN

PRACTICAL 1

INFORMATION COMMUNICATION TECHNOLOGY

ELEMENT COVERED ELEMENT 2: CREATE COMPUTERIZED DATABASE

Duration: 3 HOURS

GENERAL INSTRUCTIONS

- ✓ *Read the scenario below carefully before attempting the task.*
- ✓ *Answer all three (3) tasks. Each task is compulsory.*
- ✓ *You may use pen and paper, a whiteboard tool, draw.io, Lucidchart, or MySQL Workbench for the ERD*
- ✓ *Use MS Access for database creation*
- ✓ *Label all entities, attributes, primary keys (PK), and foreign keys (FK) clearly on your ERD.*
- ✓ *Save your work with your Admission Number as the file name and submit as instructed*
- ✓ *If you chose to DRAW ERD using any software like draw.io, print and present the diagram to the assessor*

SCENARIO

Aberdare Technical College wants to computerize its student course registration system. Each STUDENT registers for one or more COURSES. Each course can have many students registered .(a one-to-many relationship from STUDENT to COURSE).

TASK 1: DRAW AN ENTITY RELATIONSHIP DIAGRAM (ERD)

- Identify two (2) entities from the scenario
- List at least four (4) relevant attributes for each entity.
- Underline the Primary Key (PK) for each entity.
- Draw the relationship line between the two entities and label it with the relationship name
- Indicate the cardinality/multiplicity clearly using correct ERD notation (1 and M, or crow's foot notation) to show the one-to-many relationship.
- Use standard ERD symbols.

TASK 2: CREATE TABLES

- Based on your ERD, create the STUDENT table and the COURSE table in MS Access
- Each table must have a clearly defined Primary Key.
- Assign an appropriate data type and size to every column
- The many side entity should have the foreign key for creating relationship.
- Insert at least three (3) sample records into each table.

TASK 3: CREATE A ONE-TO-MANY RELATIONSHIP

- Establish a working one-to-many relationship between STUDENT and COURSE using the Foreign Key created in Task 2.

- Enforce referential integrity
- Demonstrate that the relationship works by running a query (or showing a relationships diagram) that lists each course together with all students registered under it.
- Show or export evidence of the relationship by taking a screenshot and save it in the file you created.

END OF TASK